

Software Engineering LAB 1 Assignment

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Problem Statement

- **Project:** *Automated Web Data Acquisition and Preprocessing System for LLM Training*
- Large Language Models require **large volumes of high-quality textual data** for training.
- Manually collecting web data is:
 - Time-consuming
 - Difficult to scale
 - Prone to irrelevant and noisy content
 - Likely to contain duplicate data
 - Often poorly structured
- The proposed system will:
 - Accept **seed URLs** as input.
 - Automatically crawl and discover relevant webpages.
 - Extract textual content from webpages.
 - Remove irrelevant/noisy content.
 - Detect and remove duplicate URLs and content.
 - Extract useful metadata such as URL, title, timestamp, and content hash.
 - Store the processed data in a **structured format suitable for NLP/LLM training**.

Why Agile SDLC Model?

- **Incremental development:** The crawler can be developed as separate modules such as URL management, crawling, extraction, deduplication, preprocessing, and storage.
- **Flexibility:** Webpages have different structures and unexpected issues such as broken links, dynamic content, and duplicate pages may require changes during development.
- **Continuous testing:** Each component can be tested independently and improved in subsequent iterations.
- **Early working prototype:** A basic crawler can be developed first and additional features can be added in later iterations.

Why not other SDLC models?

- **Waterfall:** Not ideal because it assumes requirements are finalized early and changes later are difficult to accommodate.
- **V-Model:** More rigid and testing are closely tied to predefined development stages, making iterative changes less convenient.
- **Spiral:** Better suited for large, high-risk projects requiring extensive risk analysis; it would add unnecessary complexity for a small academic project.